

PH102: Tutorial 11 solution

1. Suppose the current I_1 passes through the outer coil, then the magnetic field produced by it is given by

$$B_1 = \mu_0 \frac{N_1 I_1}{L_1}$$

As we have $r_2 \ll r_1$ and $L_2 \ll L_1$, we may consider the magnetic field B_1 as uniform across the inner coil. Thus the magnetic flux crossing the inner coil is

$$\begin{aligned}\Phi_{12} &= N_2 B_1 A_2 \\ &= \mu_0 \frac{N_1 N_2 I_1}{L_1} \pi r_2^2\end{aligned}$$

Thus the mutual inductance of the two coils is

$$\begin{aligned}M &= \frac{\Phi_{12}}{I_1} = \mu \frac{N_1 N_2}{L_1} \pi r_2^2 \\ &= 39.5 \mu H\end{aligned}$$

(verify yourself!!!)

2.

$$\vec{B} = \hat{x} B_0 \cos 2x \cos(\omega t - \beta y)$$

$$\vec{J}_D = \frac{d\vec{D}}{dt} = \frac{1}{\mu_0} \vec{\nabla} \times \vec{B}$$

$$= \frac{1}{\mu_0} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & 0 & 0 \end{vmatrix}$$

$$= -\hat{z} \frac{\beta B_0}{\mu_0} \cos(2x) \sin(\omega t - \beta y)$$

$$\vec{D} = \int \vec{J}_D dt$$

$$= -\hat{z} \int \left[\frac{\beta B_0}{\mu_0} \cos(2x) \sin(\omega t - \beta y) \right]$$

$$= \hat{z} \frac{\beta B_0}{\omega \mu_0} \cos(2x) \cos(\omega t - \beta y)$$

$$\rho = \vec{\nabla} \cdot \vec{D}$$

$$= \frac{\partial D_z}{\partial z}$$

$$= 0$$

3. (a)

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{E} = 0$$

$$\implies \frac{\partial \vec{B}}{\partial t} = 0 \implies \vec{B} \text{ is independent of } t$$

(b)

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\implies \int \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}; \Phi = \int \vec{B} \cdot d\vec{A}$$

$$\vec{E} = 0$$

$$\implies \frac{d\Phi}{dt} = 0 \implies \Phi \text{ through the loop is constant}$$

(c)

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

So,

$$\vec{E} = 0, \vec{B} = 0$$

$$\implies \vec{J} = 0$$

$\implies$ any current must be at the surface.

(d) A rotating shell of radius 'a' produces a uniform magnetic field inside

$$\vec{B} = \frac{2}{3} \mu_0 \sigma \omega a \hat{z}$$

In order to cancel such a field we need to have

$$\sigma \omega a = -\frac{3}{2} \frac{B_0}{\mu_0}$$

We know

$$\vec{K} = \sigma \vec{v}$$

$$= \sigma \omega a \sin \theta \hat{\phi}$$

$$\boxed{\vec{K} = -\frac{3}{2} \frac{B_0}{\mu_0} \sin \theta \hat{\phi}}$$

4. Using the cylindrical coordinate (r, θ, z) with z-axis along the axis of the solenoid we have
(a)

$$\vec{B} = \mu_0 n K t \hat{z}$$

(b)

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\Rightarrow \frac{1}{r} \left[\frac{\partial}{\partial r} (r E_\theta) - \frac{\partial E_r}{\partial \theta} \right] = -\mu_0 n L$$

By symmetry E does not depend on the θ and integrating, we obtain

$$\vec{E} = -\frac{\mu_0 n K r}{2} \hat{\theta}$$

(c)

$$\vec{S} = \vec{E} \times \vec{H} = -\frac{\mu_0 n^2 K^2}{2} r t \hat{r}$$

So the energy flows into the cylinder along the radial directions. The energy flowing in unit of time is then

$$\frac{du}{dt} = 2\pi r l S = \mu_0 V n^2 K^2 t, \text{ V is the volume of the cylinder}$$

Self-inductance per unit length of the solenoid is

$$L = \frac{n B \pi r^2}{i} = \mu_0 n \pi r^2$$

Hence,

$$\frac{d}{dt} \left(\frac{1}{2} L i^2 \right) = \mu_0 V n^2 K^2 t = \frac{du}{dt}$$

5.

$$\vec{E} = 50 e^{i(kz - 10^{15}t)} \hat{x}$$

Comparing the above equation with

$$\vec{E} = E_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{\eta}$$

We obtain that the direction of propagation is along the $\hat{z}$ direction. Angular velocity $\omega = 10^{15} s^{-1}$. Wave number $k = \frac{\omega}{c} = \frac{10^{15} s^{-1}}{3 \times 10^8 m/s} = 3.3 \times 10^6 m^{-1}$. Now using the Maxwell equation

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

We can write

$$\begin{aligned}\vec{k} \times \vec{E} &= \omega \vec{B} \\ \implies \vec{B} &= \frac{\vec{k} \times \vec{E}}{\omega} \\ &= \hat{y} \frac{k E_0}{\omega} e^{i(kz - \omega t)}\end{aligned}$$

$$\boxed{\vec{B} = \hat{y} \frac{50}{c} e^{i(kz - 10^{15}t)} \omega b / m^2}$$

6. (a)

$$\vec{B} = \hat{y} \frac{E_0}{c} \cos(kz - \omega t + \phi)$$

(b)

$$\vec{B} = \hat{y} \frac{E_0}{c} e^{i(kz - \omega t + \phi)}$$

The complex amplitude of $\vec{B}$ is

$$\vec{B}_0 = \hat{y} \frac{E_0}{c} e^{i\phi}$$

(c)

$$\begin{aligned}\vec{s} &= \vec{E} \times \vec{H} \\ &= \vec{E} \times \frac{\vec{B}}{\mu_0} = \hat{z} \frac{E_0^2}{\mu_0 c} \cos^2(kz - \omega t + \phi)\end{aligned}$$

$$\langle \vec{s} \rangle = \hat{z} \frac{E_0^2}{2\mu_0 c}$$

As

$$\langle \cos^2(kz - \omega t + \phi) \rangle = \frac{1}{2}$$

.